S.No. : 551 **BCS 3604**

No. of Printed Pages : 05

Following Paper ID and Roll No. to be filled in your Answer Book.										
PAPER ID : 33228	**Roll No.**									

B. Tech. Examination, 2024-25

(Even Semester)

COMPILER DESIGN

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION–A

1. Attempt all parts of the following : $8 \times 1 = 8$

   (a) What are the operation required on a symbol table?

   (b) Why loop optimization is so important than other code optimization?

   (c) Differentiate compile and interpreter.

   (d) Differentiate parse tree and annotated parse tree.

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(e) What is directed acyclic graph?

(f) Specify the two function that the parsing table of LR parser can have.

(g) Draw syntax tree for expression :

$$(a + b) * (a + b + c)$$

(h) What are the goal of error handler in parser?

SECTION – B

2. Attempt any two parts of the following : 2×6=12

(a) What are the phases of the compiler? Explain each phase in detail.

(b) Consider the grammar :

$$S \rightarrow icts$$
$$S \rightarrow ictses$$
$$S \rightarrow a$$
$$S \rightarrow b$$

Construct a leftmost and rightmost derivation for the string : ibtibtaea

(c) Generate three address code for the following statement assuming three registers are available:

t2 t4

$$x = a/(b + c) - d^{*} (e + f)$$

t1 t3 t2

(d) What do you understand by boot strapping? Explain with help of example.

SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. 8×5=40

3. (a) Explain about basic parsing techniques. What is top down parsing? Explain in detail.

(b) Discuss the role of syntax directed translation scheme. Describe the various code optimization technique in detail.

(c) Explain activation tree in runtime environment.

4. (a) Consider the grammar with the following translation rules & E as the start symbol :

$$E \rightarrow E1+T \ \{print ('+');\}$$
$$E \rightarrow T$$
$$T \rightarrow T1 * F \ \{print ('*');\}$$
$$T \rightarrow F$$
$$F \rightarrow num \ print ('num.val');\}$$

Construct parse tree for the string 2 + 3 * 4 find what will be printed.

(b) Construct the LL (1) parsing table for the following grammar?

E→E+T/T

T→T∗F

F→(E)/id

What happens if multiple entries occurring in your parsing table? Justify your answer.

2 (c) Differentiate between the top down parsing and bottom up parsing.

5. (a) Construct SLR parsing table for the following grammar :

E→ T+E/T R1 R2

T→ id R3

(b) Show that the following grammar is LL (1) but not SLR (1) :

S→ AaAb / BbBa

A→E /ε

B→E B→E/ε

(c) List the advantages and disadvantages of operator precedence parsing.

Construct the operator precedence parser :

E→ E+E

E→ E∗E

E→ id

6. (a) Explain DAG representation of basic blocks with an example. Discuss the strategies for loop optimization and dead code elimination.

(b) Define peephole optimization. List the characteristics of peephole optimization. Explain the main issues of code generation in detail.

(c) Explain various run time storage allocation strategies. Give the general structure of activation record. Give the purpose of each component.

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